

0 clubs; 2.5 hours (including examples)

Problem 1 (AIME 2020)

We want,

$$v_3(149^n - 2^n) \geq 3; v_5(149^n - 2^n) \geq 5; v_7(149^n - 2^n) \geq 7;$$

Using Lifting the Exponent directly on the first,

$$v_3(149^n - 2^n) = v_3(149) + v_3(n) = 1 + v_3(n) \geq 3$$

Therefore, $9 \mid n$.

For the second one, $149^n - 2^n \equiv 0 \pmod{5} \Leftrightarrow n \equiv 0 \pmod{4}$. Therefore, we only have to consider multiples of 4's. We let $n = 4k$ for some integer k .

$$v_5(149^{4k} - 2^{4k}) = v_5(149^4 - 2^4) + v_5(4k) \geq 5$$

$$v_5(149^4 - 2^4) = v_5((149 - 2)(149 + 2)(149^2 + 2^2)) = 1$$

Therefore, $4 \cdot 5^4 \mid n$.

For the last condition, we apply directly,

$$v_7(149^n - 2^n) = v_7(149 - 2) + v_7(n) \geq 7$$

Therefore, $7^5 \mid n$.

The minimum n is hence $n = 2^2 \cdot 3^2 \cdot 5^4 \cdot 7^5$.

Problem 2 (Poland)

When v_p is applied over all primes p ,

$$ab = cd \Leftrightarrow v_p(ab) = v_p(cd) \Leftrightarrow v_p(a) + v_p(b) = v_p(c) + v_p(d)$$

We wish to prove that, over all primes p ,

$$\gcd(a, c) \gcd(a, d) = a \gcd(a, b, c, d)$$

$$\Leftrightarrow \min(v_p(a), v_p(c)) + \min(v_p(a), v_p(d)) = v_p(a) + \min(v_p(a), v_p(b), v_p(c), v_p(d))$$

SKETCH: Casework it out.

4 clubs; 3 hours (including examples)

Problem 3 (Christmas JMO 2023)

We claim the only solution is $(a, b, p) = (1, 1, 2)$.

Note that $a, b \not\equiv 0 \pmod p$ since if both are divisible by p , the $p = v_p(a^p + b^p) = v_p(p!) = 1$. (!)

We hence take modulo p and use Fermat's Little Theorem,

$$a^p + b^p = p! \Leftrightarrow a + b \equiv 0 \pmod p$$

Therefore, $a + b \geq p$, meaning that, $a^p + b^p \geq 2 \left(\frac{p}{2}\right)^p$ by Jensen's, and $2 \left(\frac{p}{2}\right)^p$ is clearly greater than $p!$ for $p > 2$.

For $p = 2$, $a^2 + b^2 = 2! = 2$, so $a = b = 1$.

Problem 4 (Putnam 2003 B3)

When v_p is applied over all primes p ,

$$\prod_{k=1}^n \text{lcm}\left(1, 2, \dots, \left\lfloor \frac{n}{k} \right\rfloor\right) = n! \Leftrightarrow \sum_{k=1}^n v_p\left(\text{lcm}\left(1, 2, \dots, \left\lfloor \frac{n}{k} \right\rfloor\right)\right) = v_p(n!)$$

By Legendre's,

$$\begin{aligned} &\Leftrightarrow \sum_{k=1}^n v_p\left(\text{lcm}\left(1, 2, \dots, \left\lfloor \frac{n}{k} \right\rfloor\right)\right) = \sum \left\lfloor \frac{n}{p^i} \right\rfloor \\ &\Leftrightarrow \sum_{k=1}^n \left(i, \text{ where } i \text{ is the max where } p^i \leq \left\lfloor \frac{n}{k} \right\rfloor\right) = \sum \left\lfloor \frac{n}{p^i} \right\rfloor \\ &\Leftrightarrow \sum_{k=1}^n \left\lfloor \log_p \frac{n}{k} \right\rfloor = \sum \left\lfloor \frac{n}{p^i} \right\rfloor \end{aligned}$$

These count the same thing.

Problem 5 (Shortlist 1991)

We claim the answer is 1991.

$$v_{1991}(1990^{1991^{1992}} + 1992^{1991^{1990}}) = v_{1991}\left((1990^{1991^2})^{1991^{1990}} - (-1992)^{1991^{1990}}\right)$$

Since $1990^{1991^2} - (-1992) \equiv -1 - (-1) \equiv 0 \pmod{1991}$, we apply Lifting the Exponent,

$$= v_{1991}(1990^{1991^2} + 1992) + v_{1991}(1991^{1990}) = v_{1991}(1990^{1991^2} + 1992) + 1990$$

We now prove the first sum $v_{1991}(1990^{1991^2} + 1992) = 1$ to be done.

$$v_{1991}(1990^{1991^2} + 1992) = v_{1991}\left((1990^{1991^2} - (-1)^{1991^2}) + 1991\right)$$

Once again by LTE,

$$v_{1991}(1990^{1991^2} - (-1)^{1991^2}) = v_{1991}(1990 + 1) + v_{1991}(1991^2) = 3$$

However, $v_{1991}(1991) = 1$. Therefore,

$$v_{1991}\left((1990^{1991^2} - (-1)^{1991^2}) + 1991\right) = 1$$

10 clubs; 4.5 hours (including examples)

Problem 6 (USAMO 2023/4)

If $N = 1$, then Bob has no choices, so we are done.

We now look at $N > 1$. We let $n_1, n_2, \dots, n_N$ be the numbers, and an array $v_1, \dots, v_N$ where $v_i = v_2(n_i)$. The game ends whenever $v_1 = \dots = v_N = 0$.

When Bob moves, he can decrease v_i by 1 for some i .

For Alice's move, say she selects n_i ,

If $v_2 = v_2(a)$, then $v_2(n_i + a) > v_2(n_i)$.

Otherwise, if $v_2(n_i) \neq v_2(a)$, then $v_2(n_i + a) \leq v_2(n_i)$.

We claim that Alice can force the game to go on if and only if $v_i \geq v_2(a)$ for some i .

If $v_i \geq v_2(a)$ for some i , Alice can just add a to this n_i whenever $v_i = v_2(a)$, and add a to some other arbitrary number at the other times. Therefore, Alice can guarantee that, after her turn, $v_i > v_2(a) \geq 0$, meaning the game doesn't end.

If $v_i < v_2(a)$ for all i , Alice's moves cannot change v_i , so $\sum v_i$ will decrease by one every move Bob places until it reaches zero and the game moves.

In the latter case which we are concerned about, Bob's move does not change the outcome, so we're done.

Problem 7 (CGMO 2012/8)

$$2012 = 2^2 \cdot 503$$

We want the amount of numbers where,

$$v_{503}\left(\binom{2012}{k}\right) \geq 1; v_2\left(\binom{2012}{k}\right) \geq 2;$$

For the first condition,

$$v_{503}\left(\binom{2012}{k}\right) \geq 1 \Leftrightarrow v_{503}(2012!) - v_{503}(k!) - v_{503}((2012 - k)!) \geq 1$$

$$\Leftrightarrow 4 - v_{503}(k!) - v_{503}((2012 - k)!) \geq 1$$

Since $k < 503^2$, we use Legendre's to get $v_{503}(k!) = \left\lfloor \frac{k}{503} \right\rfloor$.

$$\Leftrightarrow \left\lfloor \frac{k}{503} \right\rfloor + \left\lfloor \frac{2012 - k}{503} \right\rfloor \leq 3$$

This holds for $k \neq 0, 503, 1006, 1509, 2012$.

For the second condition,

$$v_2\left(\binom{2012}{k}\right) \geq 2 \Leftrightarrow v_2(2012!) - v_2(k!) - v_2((2012 - k)!) \geq 2$$

By Legendre's, $v_2(n!) = n - s_2(n)$.

$$\Leftrightarrow 2012 - s_2(2012) - k + s_2(k) - 2012 + k + s_2(2012 - k) \geq 2$$

$$\Leftrightarrow s_2(k) + s_2(2012 - k) \geq 2 + s_2(2012)$$

Notice that if we are adding a, b , the number of times we carry over when adding the two is the positive difference $s_2(a) + s_2(b) - s_2(a + b)$. Hence, we must carry over at least twice in binary when adding $k, 2012 - k$.

$$2012 = 11111011100_2$$

We can complementary count.

Ways to do so w/ no carrying over:

We can count by construction how many k we have. There are 256 ways.

$$2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 1 \cdot 2 \cdot 2 \cdot 2 \cdot 1 \cdot 1 = 256$$

Ways to do so carrying over exactly once:

We casework on where we can carry over.

Carrying over 1st-5th digit: Impossible since that'll mean we carried over on the digit to the right.

Carrying over on the 6th digit: $2 \cdot 2 \cdot 2 \cdot 2 \cdot 1 \cdot 1 \cdot 2 \cdot 2 \cdot 2 \cdot 1 \cdot 1 = 128$ ways.

Carrying over 7th-9th digit: Impossible since that'll mean we carried over on the digit to the right.

Carrying over on the 10th digit: 128 ways similarly.

Carrying over on the 11th digit: Impossible since that'll mean we carried over on 10th digit too.

Therefore, there are $2013 - 512 = 1501$ solutions here following the second condition.

However, those include 503, 1006, and 1509, so the actual answer is 1498.

Problem 8 (USEMO 2020/1)

We let $a = \frac{lcm(x,y)+lcm(y,z)}{lcm(x,z)}$, and find all possible a .

We claim that all even numbers are achievable. We let $a = 2k$, plug in $x = z = 1; y = k$.

For all the odd numbers, we want $v_2(a) = 0$. We prove this is impossible using casework.

Case 1: $v_2(x) = v_2(z)$.

$$\text{Subcase 1a: } v_2(y) \geq v_2(x) = v_2(z). \quad v_2\left(\frac{lcm(x,y)+lcm(y,z)}{lcm(x,z)}\right) > 0$$

$$\text{Subcase 1b: } v_2(x) = v_2(z) > v_2(y). \quad v_2\left(\frac{lcm(x,y)+lcm(y,z)}{lcm(x,z)}\right) = 1.$$

Case 2: $v_2(x) > v_2(z)$.

$$\text{Subcase 2a: } v_2(z) \geq v_2(y). \text{ Therefore, } v_2\left(\frac{lcm(x,y)+lcm(y,z)}{lcm(x,z)}\right) = v_2(x) > v_2(y) \geq 0.$$

$$\text{Subcase 2b: } v_2(x) > v_2(y) > v_2(z). \quad v_2\left(\frac{lcm(x,y)+lcm(y,z)}{lcm(x,z)}\right) < 0$$

$$\text{Subcase 2c: } v_2(x) = v_2(y) > v_2(z). \quad v_2\left(\frac{lcm(x,y)+lcm(y,z)}{lcm(x,z)}\right) = 1.$$

$$\text{Subcase 2d: } v_2(y) > v_2(x) > v_2(z). \quad v_2\left(\frac{lcm(x,y)+lcm(y,z)}{lcm(x,z)}\right) > 0.$$

Case 3: $v_2(x) < v_2(z)$. Symmetric to Case 2.

21 clubs; 5.5 hours (including examples)

Problem 9 (BAMO 2018/4)

We want to prove that, over all primes p ,

$$v_p\left(\frac{a}{b} + \frac{b}{c} + \frac{c}{a}\right) \geq 0 \Rightarrow 3 \mid v_p(a) + v_p(b) + v_p(c)$$

Other cases are significantly easier.

Let $v_p(a) \leq v_p(b) \leq v_p(c)$.

If $v_p(a) = v_p(b)$ or $v_p(b) = v_p(c)$, then $v_p(a) = v_p(b) = v_p(c)$ which is trivial.

Therefore, consider the case where $v_p(a) < v_p(b) < v_p(c)$. This is only possible when,

$$v_p(b) - v_p(a) = v_p(c) - v_p(b)$$

In this case $v_p(a) + v_p(b) + v_p(c) = 3v_p(b)$, so $3 \mid v_p(a) + v_p(b) + v_p(c)$.

Problem 10 (Putnam 2017 B3)

Since $\frac{1}{2} = 2^{-1}$, when we have $a = f\left(\frac{1}{2}\right)$, c is just the digits right of the decimal point of a in base 2. Therefore, for a to be rational, c becomes periodic after a certain point.

$$f\left(\frac{1}{2}\right) \in \mathbb{Q} \Rightarrow c \text{ becomes periodic after a certain point.}$$

We prove that the latter condition is false to prove $f\left(\frac{1}{2}\right)$ is irrational.

For the sake of contradiction, let c becomes periodic after a certain point i and $f\left(\frac{2}{3}\right) = \frac{3}{2}$, and the period be j . We have that,

$$\sum_{n=0}^{\infty} c_n \frac{2^n}{3} = \sum_{n=0}^{i-1} c_n \frac{2^n}{3} + \left(\sum_{n=i}^{i+k} c_n \frac{2^n}{3} \right) \cdot \frac{\left(\frac{2}{3}\right)^i}{1 - \left(\frac{2}{3}\right)^k} = \frac{3}{2}$$

We take v_2 ,

$$\Rightarrow v_2\left(\sum_{n=0}^{\infty} c_n \frac{2^n}{3}\right) = v_2\left(\sum_{n=0}^{i-1} c_n \frac{2^n}{3} + \left(\sum_{n=i}^{i+k} c_n \frac{2^n}{3}\right) \cdot \frac{\left(\frac{2}{3}\right)^i}{1 - \left(\frac{2}{3}\right)^k}\right) = v_2\left(\frac{3}{2}\right)$$

We know that,

$$v_2\left(\sum_{n=0}^{i-1} c_n \frac{2^n}{3}\right) = 0, v_2\left(\sum_{n=i}^{i+k} c_n \frac{2^n}{3}\right) = i, v_2\left(\frac{\left(\frac{2}{3}\right)^i}{1 - \left(\frac{2}{3}\right)^k}\right) = i$$

Therefore, it is impossible for v_2 of the whole thing to be negative. (!)

Problem 12 (HMMT 2016 A7)

Note that,

$$A_n = A_{n-1} \cdot 10^{\lfloor \log_{10} 2^{10} \rfloor + 1} + 2^{10n}$$

The answer is n if $v_2(A_{n-1} \cdot 10^{\lfloor \log_{10} 2^{10} \rfloor + 1}) \geq 170$.

We look at v_2 of the terms. When $n = 1$,

$$v_2(A_1) = v_2(1024) = 10;$$

$$v_2(A_2) = v_2(2^{10} \cdot 10^7 + 2^{20}) = 17;$$

$$v_2(A_3) = v_2(2^{17} \cdot 10^{10} + 2^{30}) = 27;$$

$$v_2(A_4) = v_2(2^{27} \cdot 10^{13} + 2^{40}) = 41;$$

From this point on, $v_2(10^{\lfloor \log_{10} 2^{10n} \rfloor + 1}) > 10$, so $v_2(A_n) = 10n$ for $n \geq 5$. We want,

$$v_2(A_{n-1} \cdot 10^{\lfloor \log_{10} 2^{10n} \rfloor + 1}) \geq 170$$

Since, for this to be true, $n \geq 5$ obviously,

$$\Leftrightarrow 10(n-1) + \lfloor \log_{10} 2^{10n} \rfloor + 1 \geq 170$$

$$\Leftrightarrow 10(n-1) + 10n \log_{10} 2 \geq 170$$

We get that $n \geq 14$, so we're done.

Problem 13 (TSTST 2019/7)

Over all primes $p \leq 10^{100}$, we pick c_p such that $e_p = v_p(f(c_p))$ is maximal.

Note we plug in $y = c_p$ to get,

$$\begin{aligned} \min(v_p(f(x)), e_p) &= \min(v_p(x - c_p), e_p) \\ \Rightarrow v_p(f(x)) &= \min(v_p(x - c_p), e_p) \end{aligned}$$

Therefore, we can have, over all $p \leq 10^{100}$,

$$n = \prod p^{e_p}$$

By CRT, we can also select an m such that, over all $p \leq 10^{100}$,

$$m \equiv -c_p \pmod{p^{e_p}}$$

Therefore,

$$\begin{aligned} v_p(f(x)) &= \min(v_p(x - c_p), e_p) \\ \Leftrightarrow v_p(f(x)) &= \min(v_p(x - m), v_p(n)) \\ \Leftrightarrow f(x) &= \gcd(x - m, n) \end{aligned}$$

Problem 14 (TSTST 2013/8)

We claim that $f(1), f(2), \dots, f(3^n)$ leave distinct remainders under division by 3^n for all positive n . We prove this using induction.

Base case: $n = 0$. Obviously 1 is unique.

Inductive step: $n = k \rightarrow n = k + 1$.

Expanding and telescoping,

$$f(n + 3^k) - f(n) \equiv 2^{f(n)} + 2^{f(n+1)} + \dots + 2^{f(n+3^k-1)} \pmod{3^{k+1}}$$

By Euler's totient theorem, $f(n) \pmod{3^{k+1}}$ cycles every $2 \cdot 3^{k+1}$. However, since f takes only odd values, it must cycle every 3^{k+1} . Therefore,

$$\equiv 2^1 + 2^3 + \dots + 2^{2 \cdot 3^k - 1} \equiv 2 \cdot \frac{4^{3^k} - 1}{3} \pmod{3^{k+1}}$$

We perform lifting the exponent on $2 \cdot \frac{4^{3^k} - 1}{3}$.

$$v_3\left(2 \cdot \frac{4^{3^k} - 1}{3}\right) = v_3(4^{3^k} - 1) - 1 = v_3(4 - 1) + v_3(3^k) - 1 = k$$

Therefore, $f(n), f(n + 3^k), f(n + 2 \cdot 3^k)$ have different modulo 3^{k+1} s over all n , and our induction is done.

Conclusion: By mathematical induction, the hypothesis is valid for all n including $n = 2013$.

46 clubs, 9 hours